

Victoria Nguyen

682-407-4120 | victoria.n.nguyen05@gmail.com | in/vicngan | github.com/vicngan | vicnganfolio.netlify.app

EDUCATION

University of Michigan - Ann Arbor

Ann Arbor, MI

Bachelor of Science in Computer Science, Minor in Entrepreneurship

May 2028

- **Relevant Coursework:** Data Structures and Algorithms, Discrete Math, Entrepreneurship Practicum, Digital Product Design
- **Engagements:** MProduct - Product Management Team, GenAPA - Production Chair, Depression Center - Student Advisor

WORK EXPERIENCE

Michigan Medicine - Department of Computational Medicine

Ann Arbor, MI

Research Assistant - Genomic Foundation Model AI

Jan 2026 – Present

- **Architected a FastAPI middleware layer** to bridge React frontends with transformer model inference, reducing query latency for 50kb genomic windows by **40%** through optimized caching and asynchronous I/O.
- **Engineered a Conversational AI Interface** for a genomic foundation model enabling researchers to query DNA binding predictions across the **3B+ base-pair human genome**.
- **Developed an interactive D3.js Genome Browser** to visualize transcription factor binding affinities; utilized **virtualized rendering and debounced redraws** to handle datasets of **10,000+ annotations** with real-time zooming.
- Collaborated with PhD researchers to translate complex metrics into interpretability features like motif highlight maps, enabling non-technical stakeholders to explore high-dimensional model behavior.

Michigan Medicine - Department of Pediatrics

Ann Arbor, MI

Research Assistant - ABC Growing Kids Project

Apr 2025 – Present

- **Engineered automated data cleaning scripts** in **Python** to process **large-scale behavioral datasets**, standardizing formats and integrating validation checks to prevent data loss, saving 15+ hours per week and eliminated transcription errors
- **Achieved 90%+ inter-rater reliability** on video behavioral coding using **Observer XT** and **analytical skills**, creating a high-fidelity "Ground Truth" dataset for training downstream predictive models.
- **Performed quantitative analysis** on 60+ pediatric feeding experiment videos, applying **standardized observational scales** to quantify caregiver responsiveness, child hunger/satiety cues, and feeding interactions for downstream statistical analysis.

Michigan Medicine University Hospital

Ann Arbor, MI

Nurse Aide II

May 2025 – Jan 2026

- **Leveraged direct clinical experience with EPIC EMR** workflows to identify critical usability gaps, directly informing the **UI/UX architecture** of the Heartline Care Companion software project.
- **Collaborated with interdisciplinary medical professionals**, communicated clearly, escalated issues respectfully, and performed accurate handoffs in a fast-paced inpatient setting, maintaining precise, timely documentation that supported seamless patient care
- **Managed sensitive patient data entry** for a 32-bed transplant unit, including admissions, orders, vitals, and progress notes, while adhering to role-based access controls and minimum-necessary data principles.

PROJECTS

SynthLab | *React, FastAPI, Python, SDV, FAISS, RAG* |

Dec 2025 – Present

- **Designed a full-stack research platform** using **React/Vite** (frontend) and **FastAPI** (backend), enabling secure synthetic data generation for clinical trials with a modern, responsive UI.
- **Architected data synthesis pipeline** ensuring physiological constraints are enforced; critical for medical simulations, analogous to enforcing physics constraints in aerospace systems
- **Developed a Literature Intelligence** module using **FAISS vector embeddings** and LLM APIs to allow researchers to perform **semantic search and automated summarization** on local PDF libraries.

Heartline Care Companion | *Python, Streamlit, React Native, SQLite*

Oct 2025 – Nov 2025

- **Architected a cross-platform care companion** using a **Streamlit** core and an **Expo/React Native** shell, featuring an adaptive UI that flexes routines based on user energy levels.
- **Implemented advanced visualizations** using **Matplotlib** and **Altair** to render dual-line charts of mood and energy trends, providing longitudinal wellness insights over 30-day windows.
- **Integrated multi-channel notifications** via **Twilio SMS** and **SMTP** to trigger automated caregiver alerts when symptom algorithms detect a **20% deviation** from patient baselines.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript (ES6+), TypeScript, SQL, HTML/CSS, Bash

Frameworks: React, FastAPI, Vite, Node.js, TailwindCSS, Express, React Native

Data & AI: Pandas, Synthetic Data Vault, FAISS (Vector Search), RAG, Plotly, D3.js, NumPy, AI/LLM, Google Cloud

Tools & Infrastructure: Git/GitHub, Docker, MongoDB, REST APIs, Jupyter, Figma, Microsoft/Google Workspace, Excel